

Conference-Revision Supplement

Claim-preserving editorial material for integration into the editable manuscript

Purpose. This supplement improves presentation and records the sections required for a conference revision without changing the paper's algorithm, contributions, datasets, or reported numerical results. It is appended because the available manuscript is PDF-only; no editable source was supplied.

Editorial actions for the source manuscript

- Apply one term consistently: “predicate-aware graph traversal” (GRAPH).
- Define every symbol before its first use and retain Table 2 as the notation authority.
- Replace unfinished cross-references such as “§??” with a resolved section reference before submission.
- Repair the malformed illustrative query in the abstract and motivating example from the source; its omitted predicate text cannot be reconstructed safely from the PDF alone.
- Use sentence-case captions that state the metric, workload, backend, and whether values are means, medians, or confidence intervals.
- Keep reported prototype results separate from planned million-scale evaluations.

Figure Placement and Captions

Insert Figure A after §4.1 (Architecture Overview) and Figure B at the beginning of §6 (Evaluation). Both diagrams are vector PDFs and use a consistent blue/green palette, aligned arrows, and readable conference-paper typography.

Suggested captions

Figure A. SDAO adaptive-planning workflow. Structured-predicate and vector-density features feed the estimator and cost model; the planner selects one physical strategy and logs feedback for later estimator updates.

Figure B. Evaluation pipeline. The dataset and generated query workload are processed by the estimator and planner, executed under a selected strategy, measured, and returned to the feedback path.

6.x Ablation Study (Planned Evaluation)

Experiment design. Hold the dataset, query set, ANN parameters, and target recall constant. Compare the full SDAO configuration with one component removed at a time. Report mean and median latency, filtered Recall@10, vectors scanned, planning latency, and memory overhead. Use the same query seeds across variants.

Variant	Purpose	Latency	Recall@10	Scans	Status
Full SDAO	Reference configuration	To be measured	To be measured	To be measured	Planned
Without estimator	Static/selectivity based	To be measured	To be measured	To be measured	Planned
Without online learning	Iterative feedback updates	To be measured	To be measured	To be measured	Planned
Without GRAPH	Assess graph action	To be measured	To be measured	To be measured	Planned
Without ISE	Assess intersection action	To be measured	To be measured	To be measured	Planned
Without adaptive planning	Fixed-strategy control	To be measured	To be measured	To be measured	Planned

Results will be reported after implementation. This table is intentionally a template: no values are inferred from existing experiments.

Memory, Planner, and Statistical Evaluation

Memory Overhead

Measure resident and peak memory separately for the estimator parameters, per-attribute histograms, planner state, online training buffer, and feature cache. Report both absolute bytes and incremental bytes over the vector-index baseline at each dataset scale. Memory measurements are left for future evaluation.

Planner Overhead

Instrument feature extraction, selectivity estimation, cost evaluation, strategy decision, and total optimizer overhead with monotonic timers. Report warm and cold-cache distributions. Planner overhead will be evaluated in future work.

Statistical Evaluation Plan

For planned evaluations, use multiple independent seeds and report 95% confidence intervals. Compare paired query outcomes with a paired t-test when normality is supported; otherwise use a Wilcoxon signed-rank test. Apply a stated multiple-comparison correction where several strategies are tested. No p-values or significance claims are made until these measurements are collected.

Threats to Validity and Future Evaluation

Threats to Validity

Internal validity: the current workload is synthetic and the system is a prototype; parameter tuning may affect outcomes. External validity: the design is discussed in the context of PostgreSQL and limited datasets, so generalization to other engines and data distributions remains to be evaluated. Construct validity: Recall@10 and latency capture important but incomplete behavior; workload diversity, tail latency, update cost, and resource isolation require additional study.

Future Work

Planned evaluations include million-scale datasets, real PostgreSQL integration, direct pgvector and ACORN comparisons, cloud deployment, distributed execution, join optimization, and adaptive threshold learning. These are evaluation and implementation plans, not completed results.

Related-work framing

Position SDAO as complementary to learned-optimization systems (Bao, Neo, NeuroCard), integrated vector systems (pgvector and PostgreSQL-V), and filtered ANN systems (ACORN, NaviX, SERF, and iRangeGraph). The distinction is the proposed selectivity-conditioned, adaptive strategy-selection layer; avoid implying head-to-head performance unless directly measured.

Revision Record and Remaining Submission Risks

Changes supplied in this revision package

- Two publication-quality vector diagrams with consistent visual language.
- A claim-preserving ablation-study template and methodology.
- Explicit memory, planner-overhead, and statistical-evaluation plans.
- A threats-to-validity section and expanded future-evaluation roadmap.
- A source-level editorial checklist for consistency, captions, notation, and unresolved references.

Remaining weaknesses requiring future experiments

The PDF-only source prevents a safe line-by-line prose rewrite, bibliographic reformatting, and replacement of malformed source text. The existing paper must retain its reported results; production-scale comparisons, direct external-baseline experiments, memory measurements, planning-overhead measurements, ablations, and statistical tests remain to be performed before a final conference submission.